Finiteness of Class Groups and the Minkowski Theorem

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Abstract

This paper develops the algebraic number theory and geometry of numbers needed to prove the finiteness of ideal class groups of number fields. It introduces the basic language of algebraic integers, Dedekind domains, ideals, norms, and discriminants, then turns to lattices and the Minkowski embedding to connect number theory with geometry. Using Minkowski's lattice point theorem, the paper derives bounds on the norms of ideals representing ideal classes and concludes that the class group of a number field is finite.

1 Introduction

The ring of integers $\mathcal{O}_K$ of an algebraic number field K is the natural generalization of $\mathbb{Z}$ to higher dimensions. While it mimics the integers in many ways the failure of unique factorization is a central problem. In $\mathbb{Z}$ every number has a unique decomposition into primes. In $\mathcal{O}_K$, irreducible elements do not always behave like prime elements. For example if $K = \mathbb{Q}(\sqrt{-5})$, the ring of integers is $\mathbb{Z}[\sqrt{-5}]$. In this ring, the number 6 has two distinct factorizations into irreducibles:

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

This failure of unique factorization breaks the fundamental tools that we use to solve Diophantine equations. For example, Fermat's Last Theorem fails for rings that are not unique factorization domains because it can no longer be assumed that if $ab = c^n$ then a and b are n-th powers. Also, without a UFD, the greatest common divisor is no longer well-defined for elements, making basic modular arithmetic unpredictable.

Despite this unpredictability, the ring of integers is a Dedekind Domain which restores unique factorization for ideals even if it fails for individual numbers. Thus, we can still do arithmetic provided we work with ideals as opposed to individual elements. The ideal class group of a number ring and its order tells us the extent to which $\mathcal{O}_K$ fails to be a unique factorization domain.

2 Algebraic Number Theory Preliminaries

To get started we need a few basic number theory definitions and notations explained.

Definition 2.1. Let K be an algebraic number field (a finite extension of $\mathbb{Q}$). The **ring of algebraic integers** $\mathcal{O}_K$, also called a **number ring**, consists of all elements $\alpha \in K$ that are roots of some monic polynomial with integer coefficients:

$$f(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0 \in \mathbb{Z}[x]$$

$\mathcal{O}_K$ is the set of all elements in K that may not look like integers but they "act" like integers. The standard integers $\mathbb{Z}$ form a ring; you can only leave $\mathbb{Z}$ by dividing. $\mathcal{O}_K$ behaves the same way and as we'll see later, ideals in $\mathcal{O}_K$ also factor uniquely into prime ideals, similar to how integers factor uniquely into prime numbers.

Let's look at the number ring of $K = \mathbb{Q}(\sqrt{5})$. For sure $\mathbb{Z}[\sqrt{5}] = \{a + b\sqrt{5} | a, b \in \mathbb{Z}\}$ is in $\mathcal{O}_K$. There exists a monic polynomial in $\mathbb{Z}[x]$ for every $x \in \mathbb{Z}[\sqrt{5}]$:

$$x = a + b\sqrt{5} \implies x^2 - 2ax + a^2 - 5b^2 = 0$$

However, $\frac{1+\sqrt{5}}{2}$ is also in $\mathcal{O}_K$:

$$x = \frac{1+\sqrt{5}}{2} \implies 2x - 1 = \sqrt{5} \implies 4x^2 - 4x + 1 = 5 \implies x^2 - x - 1 = 0$$

So, $\mathcal{O}_K = \mathbb{Z}[\frac{1+\sqrt{5}}{2}]$.

Definition 2.2. Let K be a number field of degree n over $\mathbb{Q}$. Let $\sigma_1, \sigma_2, \ldots, \sigma_n$ denote the n embeddings of K in $\mathbb{C}$ that fix $\mathbb{Q}$. The **discriminant** of the set $\{\theta_1, \theta_2, \ldots, \theta_n\}$, denoted $\text{disc}(\theta_1, \theta_2, \ldots, \theta_n)$, is defined as

$$\text{disc}(\theta_1, \theta_2, \ldots, \theta_n) = \det[\sigma_j(\theta_i)]^2 = \det \begin{bmatrix} \sigma_1(\theta_1) & \sigma_2(\theta_1) & \ldots & \sigma_n(\theta_1) \\ \sigma_1(\theta_2) & \sigma_2(\theta_2) & \ldots & \sigma_n(\theta_2) \\ \ldots & \ldots & \ldots & \ldots \\ \sigma_1(\theta_n) & \sigma_2(\theta_n) & \ldots & \sigma_n(\theta_n) \end{bmatrix}^2$$

When referring to the discriminant of a basis of $\mathcal{O}_K$ as opposed to the discriminant of an element(s), the discriminant is referred to as the **field discriminant** of K and denoted d_K. Furthermore the discriminant is independent of the choice of integral basis. The discriminant gives us important information about how ideals factor in the number ring. It also can be used as a tool to verify if a set of elements form an integral basis for $\mathcal{O}_K$. It should also be noted that for an ideal I of $\mathcal{O}_K$, the discriminant is

$$\text{disc}(I) = (\mathcal{O}_K : I)^2 d_K$$

where $(\mathcal{O}_K : I)$ is the order of the factor group $\mathcal{O}_K/I$ or equivalently, the absolute norm of the ideal I.

While every number ring is an integral domain, it's generally not a unique factorization domain. This is a fairly large failure as it means elements in $\mathcal{O}_K$ might not factor uniquely. However, $\mathcal{O}_K$ has its own unique structure; every number ring is what's known as a Dedekind Domain.

Definition 2.3. An integral domain, R, is called a **Dedekind Domain** when

(i) Every ideal of R is finitely generated

(ii) Every non-zero prime ideal is maximal

(iii) R is integrally closed in its field of fractions

$$K = \{\frac{\alpha}{\beta} | \alpha, \beta \in R, \beta \neq 0\}$$

i.e. the only elements $\frac{\alpha}{\beta} \in K$ that are integral over R are $\frac{\alpha}{\beta} \in R$.

Thus, while every element in $\mathcal{O}_K$ might not factor uniquely, every ideal in a number ring does. Every ideal in a Dedekind Domain factors uniquely into a product of prime ideals similar to how every integer factors uniquely into a product of primes by the Fundamental Theorem of Arithmetic.

Now before we can define the ideal class group, fractional ideals need to be defined.

Definition 2.4. Let $\mathcal{O}$ be a Dedekind Domain and K is its field of fractions. A **fractional ideal** of K is a set of the form $\alpha I, \alpha \in K, \alpha \neq 0$, where I is a non-zero ideal of $\mathcal{O}$.

A fractional ideal is essentially an ideal that has been scaled by an element of the field, allowing it to have denominators. An integral ideal is a subset of the ring $\mathcal{O}_K$ and a fractional ideal is a subset of the field K that is an integral ideal in $\mathcal{O}_K$ once the denominators have been cleared.

Consider the integral ideal $I = \langle 2, 1+\sqrt{-5}\rangle$ in $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ where $K = \mathbb{Q}(\sqrt{-5})$. This is the set of all elements $2\alpha + (1+\sqrt{-5})\beta$ where $\alpha, \beta \in \mathcal{O}_K$. A fractional ideal, J can be created by scaling I by $\frac{1}{2}$:

$$J = \frac{1}{2}I = \left\{\alpha + \frac{1+\sqrt{-5}}{2}\beta \mid \alpha, \beta \in \mathcal{O}_K\right\}$$

Also, note that integral ideals are also fractional ideals; the denominator is just 1.

Definition 2.5. Let K be an algebraic number field. Let J_K be the group of non-zero fractional ideals of $\mathcal{O}_K$. Let P_K be the subgroup of principal fractional ideals of J_K. The **ideal class group** of K is denoted Cl_K and is the factor group of J_K/P_K.

Another way to understand this is that the ideal class group of a number ring consists of equivalence classes of nonzero ideals under the relation

$$I \sim J \text{ iff } \alpha I = \beta J \text{ for some nonzero } \alpha, \beta \in \mathcal{O}_K.$$

Let's look at $K = \mathbb{Q}(\sqrt{-5})$ and $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ again. P_K consists of every ideal that can be written as $\langle\alpha\rangle$ for some $\alpha \in K$. For example

$$\langle 1\rangle, \quad \langle 2\rangle, \quad \langle 3\rangle, \quad \langle 1+\sqrt{-5}\rangle, \quad \langle\frac{1}{2}\rangle, \text{ etc.}$$

J_K consists of all fractional ideals of $\mathcal{O}_K$ including P_K

$$\langle 1\rangle, \quad \langle 1+\sqrt{-5}\rangle, \quad \langle\frac{1}{2}\rangle, \quad \langle 2, 1+\sqrt{-5}\rangle, \quad \langle 6, 3+3\sqrt{-5}\rangle, \text{ etc.}$$

All the principal ideals are equivalent and all the other fractional ideals are equivalent if they differ by a principal ideal. Looking at the last two examples of fractional ideals we can see that they are actually equivalent over Cl_K:

$$\langle 6, 3+3\sqrt{-5}\rangle = \langle 3\rangle\langle 2, 1+\sqrt{-5}\rangle$$

since they differ by the principal ideal $\langle 3\rangle$.

In a Dedekind Domain, the property of being a Principal Ideal Domain (PID) and a Unique Factorization Domain (UFD) are equivalent. So the order of the ideal class group tells us how far away $\mathcal{O}_K$ is from being a UFD.

Definition 2.6. The order of the ideal class group of K, denoted h_K is called the **class number** of K and denotes the number of elements in Cl_K. Equivalently h_K is the number of cosets of P_K in J_K otherwise known as the **index**: $(G : H)$.

So if $h_K = 1$ it follows immediately that $\mathcal{O}_K$ must be a PID and so in turn must be a UFD because every ideal of $\mathcal{O}_K$ is a principal ideal.

Definition 2.7. Let K be a number field of degree n over $\mathbb{Q}$ and let $P \subseteq \mathcal{O}_K$ be a prime ideal. Let p be the unique rational prime contained in P. Then $\mathcal{O}_K/P$ is a finite field extension of $\mathbb{Z}_p$; the degree of this extension is called the inertial degree of P over p and is denoted $f(P|p)$. That is,

$$f(P|p) = (\mathcal{O}_K/P : \mathbb{Z}_p).$$

3 Geometry of Numbers

Before we get into proving the finiteness of the ideal class group, we need some geometry background to be able to apply Minkowski's theorems.

Definition 3.1. Let $\{v_1, v_2, \ldots, v_m\}$ be a set of m linearly independent vectors in $\mathbb{R}^n$, a **lattice** Γ is the set of all integer linear combinations of those basis vectors:

$$\Gamma = \{\sum_{i=1}^{m} a_i v_i | a_i \in \mathbb{Z}\}$$

The elements of Γ are called **lattice points**. The lattice is called **complete** if $m = n$. The **volume** of the **fundamental mesh** of Γ is

$$vol(\Gamma) = |\det(A)|$$

where

$$A = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \ldots & v_m \\ | & | & & | \end{bmatrix}$$

A simple example of a lattice is the standard integer lattice $\mathbb{Z}^2$, which forms a grid of all integer pairs in 2D space. In this case $m = 2$ so the lattice is complete and we use the standard basis $v_1 = (1, 0)$ and $v_2 = (0, 1)$. See Figure 1 for a visual. The volume of the fundamental mesh of $\mathbb{Z}^2$ is simple to calculate:

$$vol(\mathbb{Z}^2) = |\det(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix})| = |1 - 0| = 1$$

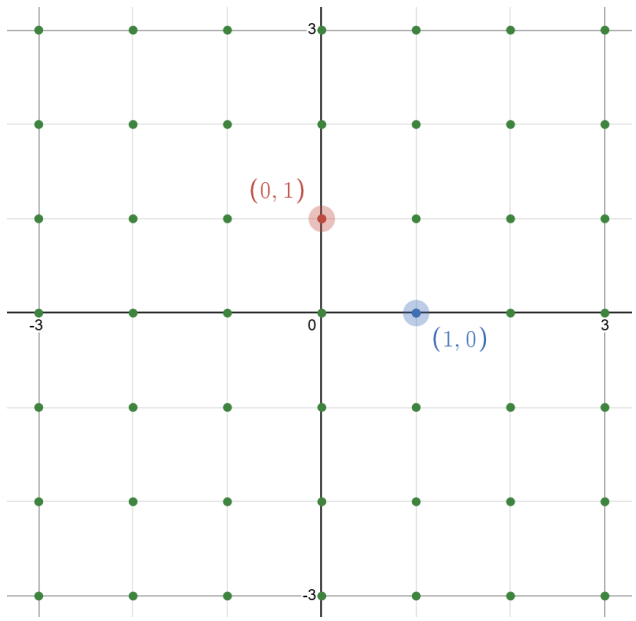

Figure 1: The $\mathbb{Z}^2$ lattice in 2D space

Definition 3.2. A subset S of $\mathbb{R}^n$ is said to be **convex** if

$$t\beta + (1 - t)\gamma \in S \quad \forall \beta, \gamma \in S \text{ and } \forall t \in [0, 1]$$

Visually, we can think of a set of points as a shape that is convex when a line connecting 2 points in the shape is also entirely in the shape. For example, a circle, square or ellipse is convex but a star or crescent moon is not. You could draw a line between 2 points on either "arm" of the star and the line would pass outside of the shape.

Definition 3.3. A subset S of $\mathbb{R}^n$ is said to be **centrally symmetric** if

$$-\alpha \in S \quad \forall \alpha \in S$$

In other words, a shape is centrally symmetric if it looks exactly the same when rotated 180°around the origin.

Theorem 3.4. (Minkowski's Lattice Point Theorem[2]). Let Γ be a complete lattice in $\mathbb{R}^n$, and let X be a centrally symmetric, convex subset of $\mathbb{R}^n$. Suppose that

$$vol(X) > 2^n vol(\Gamma).$$

Then X contains at least one nonzero lattice point $\gamma \in \Gamma$.

Let's look at the ring of Gaussian integers, $\mathbb{Z}[i]$, which can be viewed as the standard lattice $\Gamma = \mathbb{Z}^2$ in $\mathbb{R}^2$. Here the basis vectors are $v_1 = (1, 0)$ and $v_2 = (0, 1)$ and the volume is $vol(\Gamma) = 1$. Let X be a disk centered at the origin with radius R. This shape is convex and centrally symmetric with a volume (area in this case) of $vol(X) = \pi R^2$. Using Minkowski's theorem:

$$vol(X) > 2^n vol(\Gamma) \implies \pi R^2 > 2^2 * 1 \implies R > \sqrt{\frac{4}{\pi}} \approx 1.128.$$

So if we choose a radius just slightly larger than 1.128, Minkowski's theorem guarantees a non-zero Gaussian integer exists within that circle.

4 Proof of Class Group Finiteness

Now we are able to apply the theory of lattices to the study of algebraic number fields of degree n. The field can be thought of as an n dimensional vector space and we will show that $\mathcal{O}_K$ forms a lattice in the vector space. First, remember that embeddings map generators to a root of its respective minimal polynomial. A real embedding maps from $K \to \mathbb{R}$ while a complex embedding maps from $K \to \mathbb{C}$.

Take for example, the number field $K = \mathbb{Q}(\sqrt[3]{2})$. The minimal polynomial is $f(x) = x^3 - 2$ and the roots are $\theta_1 = \sqrt[3]{2}, \quad \theta_2 = \omega\sqrt[3]{2} \quad \theta_3 = \omega^2\sqrt[3]{2}$ where $\omega = e^{2\pi i/3}$. Here there is one real embedding (ρ_1) and two complex embeddings(σ_1, σ_2):

$$\rho_1(\sqrt[3]{2}) = \sqrt[3]{2}, \quad \sigma_1(\sqrt[3]{2}) = \omega\sqrt[3]{2}, \quad \sigma_2(\sqrt[3]{2}) = \omega^2\sqrt[3]{2}.$$

Definition 4.1. Let K be a number field of degree $n = r + 2s$, with r real embeddings $\rho_1, \ldots, \rho_r$ and s pairs of complex conjugate embeddings $\sigma_1, \bar{\sigma}_1, \ldots, \sigma_s, \bar{\sigma}_s$. The **canonical embedding** (or Minkowski embedding) is the injective ring homomorphism $j : K \to \mathbb{R}^n$ defined by:

$$j(\alpha) = (\rho_1(\alpha), \ldots, \rho_r(\alpha), \mathrm{Re}(\sigma_1(\alpha)), \mathrm{Im}(\sigma_1(\alpha)), \ldots, \mathrm{Re}(\sigma_s(\alpha)), \mathrm{Im}(\sigma_s(\alpha)))$$

The target space $\mathbb{R}^n$ is called the **Minkowski space**, $K_{\mathbb{R}}$.

So, for $K = \mathbb{Q}(\sqrt[3]{2})$ and $\alpha = \sqrt[3]{2}$, $r = 1$, $s = 1$ since σ_1 and σ_2 are a single conjugate pair. So, $n = r + 2s = 3$, meaning the Minkowski space is $\mathbb{R}^3$. Furthermore, since

$$\sigma_1(\sqrt[3]{2}) = \omega\sqrt[3]{2} = \sqrt[3]{2}(-\frac{1}{2} + i\frac{\sqrt{3}}{2})$$

we get

$$j(\alpha) = (\sqrt[3]{2}, -\frac{\sqrt[3]{2}}{2}, \frac{\sqrt[3]{2}\sqrt{3}}{2}).$$

It will soon be useful to see the relationship between the complex vectors and the real vectors in the Minkowski space. It turns out $|\det(M)| = 2^{-s}|\det(A)|$ where M is the matrix formed by the real vectors and A is the matrix formed by the complex vectors. For example,

$$\begin{bmatrix} \rho_1(\alpha) \\ \mathrm{Re}(\sigma_1(\alpha)) \\ \mathrm{Im}(\sigma_1(\alpha)) \end{bmatrix} = \begin{bmatrix} \sqrt[3]{2} \\ -\frac{\sqrt[3]{2}}{2} \\ \frac{\sqrt[3]{2}\sqrt{3}}{2} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2i} & -\frac{1}{2i} \end{bmatrix}}_{T} \begin{bmatrix} \rho_1(\alpha) \\ \sigma_1(\alpha) \\ \sigma_2(\alpha) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2i} & -\frac{1}{2i} \end{bmatrix} \begin{bmatrix} \sqrt[3]{2} \\ \sqrt[3]{2}(-\frac{1}{2} + i\frac{\sqrt{3}}{2}) \\ \sqrt[3]{2}(-\frac{1}{2} - i\frac{\sqrt{3}}{2}) \end{bmatrix}$$

$$|\det(T)| = |1 \cdot \det \begin{pmatrix} 1/2 & 1/2 \\ 1/2i & -1/2i \end{pmatrix}| = |-\frac{1}{4i} - \frac{1}{4i}| = |-\frac{1}{2i}| = |\frac{i}{2}| = \frac{1}{2} = 2^{-1}$$

where $s = 1$ because there is one pair of complex conjugate embeddings.

Now that we can view K as an n-dimensional euclidean space, we can interpret $\mathcal{O}_K$ and its ideals as lattices in the Minkowski space using the following lemma.

Lemma 4.2. [2] Let K be a finite extension of $\mathbb{Q}$, and let $I \neq 0$ be an ideal of $\mathcal{O}_K$. Let j be the map from K into the Minkowski space. Then $\Gamma = j(I)$ is a complete lattice in $K_{\mathbb{R}}$, and its fundamental mesh has volume

$$vol(\Gamma) = 2^{-s}\sqrt{|d_K|}(\mathcal{O}_K : I).$$

Proof. Let $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ be a $\mathbb{Z}$-basis of I. Then the lattice $\Gamma = j(I)$ is spanned by the real vectors $\{j(\alpha_1), j(\alpha_2), \ldots, j(\alpha_n)\}$ in $\mathbb{R}^n$. Let M be the $n \times n$ matrix whose rows are these vectors. Then the volume of the fundamental mesh is given by $vol(\Gamma) = |\det(M)|$. Let $\{\tau_1, \tau_2, \ldots, \tau_n\}$ be the embeddings of K into $\mathbb{C}$. Then, if

$$A = \begin{bmatrix} \tau_1(\alpha_1) & \tau_2(\alpha_1) & \ldots & \tau_n(\alpha_1) \\ \tau_1(\alpha_2) & \tau_2(\alpha_2) & \ldots & \tau_n(\alpha_2) \\ \ldots & \ldots & \ldots & \ldots \\ \tau_1(\alpha_n) & \tau_2(\alpha_n) & \ldots & \tau_n(\alpha_n) \end{bmatrix}$$

the transformation from $\mathbb{C}^n$ to $\mathbb{R}^n$ is

$$vol(\Gamma) = |\det(M)| = 2^{-s}|\det(A)|.$$

But we know the discriminant is

$$\operatorname{disc}(I) = d(\alpha_1, \alpha_2, \ldots, \alpha_n) = \det(A)^2 = (\mathcal{O}_K : I)^2 d_K.$$

So,

$$vol(\Gamma) = 2^{-s}|\det(A)| = 2^{-s}\sqrt{|d_K|}(\mathcal{O}_K : I),$$

as desired. □

In order to apply lattices of ideals to the class number, we need to first prove a theorem that will aid us to find a "bound" for ideals. The theorem comes as a direct result of the Minkowski Lattice Point Theorem. Essentially, it says that there exists a non-zero α in an ideal I such that any embedding of α is less than some chosen constants c_τ, provided that the product of these constants is large enough to guarantee that the space they define contains at least one non-zero point of the ideal's lattice. Formally,

Theorem 4.3. [2] Let $K/\mathbb{Q}$ be a finite extension, and let $I \neq 0$ be an ideal of $\mathcal{O}_K$. Let $c_\tau > 0$, for $\tau \in \operatorname{Hom}(K, \mathbb{C})$, be a real number such that $c_\tau = c_{\bar{\tau}}$ and

$$\prod_\tau c_\tau > (2/\pi)^s \sqrt{|d_K|}(\mathcal{O}_K : I).$$

Then there exists a nonzero $\alpha \in I$ such that

$$|\tau(\alpha)| < c_\tau \text{ for all } \tau \in \operatorname{Hom}(K, \mathbb{C}).$$

Proof. Let

$$X = \{(z_\tau) \in K_{\mathbb{R}} | |z_\tau| < c_\tau\}.$$

This is simply some solid shape in $\mathbb{R}^n$ bounded by some chosen constants for each embedding. This set is centrally symmetric because $|z_\tau| = |-z_\tau|$, and it's convex since for any $t \in [0,1]$ and $|z_\tau|, |w_\tau| < c_\tau$, by the triangle inequality

$$|tz_\tau + (1-t)w_\tau| \leq t|z_\tau| + (1-t)|w_\tau| < tc_\tau + (1-t)c_\tau = c_\tau.$$

Because $K_{\mathbb{R}}$ is created from r real lines and s complex planes, each point in X is the product of a real embedding (ρ) – $|z_\rho| < c_\rho$ defines a 1D interval on a real line whose volume (length) is $2c_\rho$ – and a complex pair of embeddings $(\sigma, \bar{\sigma})$ – $|z_\sigma| < c_\sigma$ defines a 2D disk in a complex plane whose volume (area) is πc_σ^2.

Then to find the total volume of X, we need to multiply all the individual volumes together. Since there are r real embeddings, we get $(2c_{\rho_1})(2c_{\rho_2})\ldots(2c_{\rho_r}) = 2^r \prod c_\rho$. And there are s complex embedding pairs, we get $(\pi c_{\sigma_1}^2)(\pi c_{\sigma_2}^2)\ldots(\pi c_{\sigma_s}^2) = \pi^s \prod c_\sigma^2$. So the total volume is

$$vol(X) = (2^r \prod c_\rho) \cdot (\pi^s \prod c_\sigma^2) = 2^r \pi^s \prod_\tau c_\tau.$$

But, from our definition of c_τ,

$$\begin{aligned} vol(X) &= 2^r \pi^s \prod_\tau c_\tau \\ &> 2^r \pi^s \left(\frac{2}{\pi}\right)^s \sqrt{|d_K|}(\mathcal{O}_K : I) \\ &= 2^{r+s}\sqrt{|d_K|}(\mathcal{O}_K : I) \\ &= 2^{r+2s}\left(2^{-s}\sqrt{|d_K|}(\mathcal{O}_K : I)\right) && \text{by Lemma 4.2} \\ &= 2^n vol(\Gamma) \end{aligned}$$

Thus, by the Minkowski Lattice Point Theorem, there exists a lattice point $j(\alpha) \in X, \alpha \neq 0, \alpha \in I$. So based on how we defined X, $|\tau(\alpha)| < c_\tau$, as desired. □

Lemma 4.4. [2] Let $K = \mathbb{Q}(\theta)$ be an algebraic number field of degree $n = r + 2s$, where θ has r real conjugates and s pairs of non real complex conjugates. Let A be an integral ideal of $\mathcal{O}_K$. Then there exists an element $\alpha(\neq 0) \in A$ such that

$$|N_{K/\mathbb{Q}}(\alpha)| \leq (\frac{2}{\pi})^s \sqrt{|d_K|}\mathfrak{N}(A).$$

Proof. Given $\epsilon > 0$, we can choose positive real numbers c_τ, for $\tau \in \mathrm{Hom}(K, \mathbb{C})$, such that $c_\tau = c_{\bar{\tau}}$ and

$$\prod_\tau c_\tau = (\frac{2}{\pi})^s \sqrt{|d_K|}\mathfrak{N}(A) + \epsilon.$$

Then by Theorem 4.3, we find an element $\alpha \in A, \alpha \neq 0$, satisfying $|\tau(\alpha)| < c_\tau$. Thus

$$|N_{K/\mathbb{Q}}(\alpha)| = \prod_\tau |\tau(\alpha)| < (\frac{2}{\pi})^s \sqrt{|d_K|}\mathfrak{N}(A) + \epsilon.$$

Since $|N_{K/\mathbb{Q}}(\alpha)|$ is a positive integer,

$$|N_{K/\mathbb{Q}}(\alpha)| \leq (\frac{2}{\pi})^s \sqrt{|d_K|}\mathfrak{N}(A).$$

□

Now we can finally prove the theorem we've been building up to.

Theorem 4.5. [2] Let $K/\mathbb{Q}$ be a finite extension. The class number $h_K = (J_K : P_K)$ is finite.

Proof. It's important to note that for every prime $p \in \mathbb{Z}$, $p\mathcal{O}_K = \langle p \rangle = P_1^{v_1} P_2^{v_2} \dots P_k^{v_k}$ where P_i is a prime ideal of $\mathcal{O}_K$ that contains p. As such, there are finitely many such ideals as $P_i \mid \langle p \rangle$. Also the absolute norm of every prime ideal is bounded by definition 2.7: $\mathfrak{N}(P_i) = p^f$ where $f = (\mathcal{O}_K/P_i : \mathbb{Z}_p)$. Because there's a finite number of prime ideals containing p, there's a finite number of prime ideals bounded by p^f for every prime integer. So choose $M = (\frac{2}{\pi})^s \sqrt{|d_K|}$. There then must be a finite number of integral ideals, A, such that

$$\mathfrak{N}(A) = \mathfrak{N}(P_1)^{v_1} \cdots \mathfrak{N}(P_r)^{v_r} \leq M.$$

So if we show that every ideal class in Cl_K has a representative with absolute norm less than M then there must be finite ideal classes and as such Cl_K would have a finite order.

Let $[A]$ be an arbitrary class in Cl_K and let A be an arbitrary representative. A is a fractional ideal so there exists some nonzero $\gamma \in \mathcal{O}_K$ such that $B = \gamma A^{-1} \subseteq \mathcal{O}_K$, i.e. B is an integral ideal of O_K. Then by Lemma 4.4 there exists $\alpha \in B$ such that

$$|N_{K/\mathbb{Q}}(\alpha)| \leq (\frac{2}{\pi})^s \sqrt{|d_K|}\mathfrak{N}(B)$$

$$\implies |N_{K/\mathbb{Q}}(\alpha)|\mathfrak{N}(B)^{-1} \leq (\frac{2}{\pi})^s \sqrt{|d_K|}$$

$$\implies \mathfrak{N}(\alpha B^{-1}) \leq (\frac{2}{\pi})^s \sqrt{|d_K|}$$

because $|N_{K/\mathbb{Q}}(\alpha)| = \mathfrak{N}(\langle \alpha \rangle)$. Then since $\alpha B^{-1} = \alpha(\gamma A^{-1})^{-1} = \alpha(\gamma^{-1}A) = \frac{\alpha}{\gamma}A$, αB^{-1} must also be an ideal of $[A]$ because it differs by an element in K. Thus since $[A]$ was chosen arbitrarily, every class in Cl_K contains an ideal with absolute norm bounded by M, and thus h_K must be finite as desired. □

So now we know that the ideal class group is finite, but how do we calculate class group and its order? Using our proof for group finiteness, we can come up with the following algorithm:[1] Given an algebraic number field K = $\mathbb{Q}(\theta)$,

1. Determine $n = [K : \mathbb{Q}]$
2. Determine r the number of real conjugates of θ. Then $s = \frac{1}{2}(n - r)$.
3. Determine d_K.
4. Compute the Minkowski bound $M_K = (\frac{2}{\pi})^s \sqrt{|d_K|}$.

5. Determine all rational primes $p \leq M_K$.

6. Determine the prime ideal factorization of each principal ideal $\langle p \rangle$ in Step 5.

7. Determine all products of these prime ideals having norm $\leq M_K$.

8. Determine the generators of Cl_K from the classes of these products.

Let's use this algorithm to find the class group for $K = \mathbb{Q}(\sqrt{-14})$. We know $n = 2$ because the minimal polynomial is $x^2 + 14$. There's only 2 roots and neither are real so $r = 0$ and $s = 1$. The field discriminant is $d_K = -56$. Thus the Minkowski bound is

$$M_K = (\frac{2}{\pi})^s \sqrt{|d_K|} = (\frac{2}{\pi})\sqrt{56} < 5.$$

The rational primes $p \leq M_K$ are $p = 2$ and $p = 3$. Using Dedekind's theorem we have

$$\langle 2 \rangle = (\langle 2, \sqrt{-14} \rangle)^2 = P_1^2, \quad \langle 3 \rangle = \langle 3, 1 + \sqrt{-14} \rangle \langle 3, 1 - \sqrt{-14} \rangle = P_2 P_3.$$

The norms of these ideals are

$$\mathfrak{N}(P_1) = 2, \quad \mathfrak{N}(P_2) = \mathfrak{N}(P_3) = 3$$

which tells us that each of these prime ideals are candidates for class representatives because their norm is less than M_K. But there could possibly be other non-prime ideals whose norm is $\leq M_K$. These ideals would all be products of P_1, P_2 or P_3. But,

$$\mathfrak{N}(P_1)\mathfrak{N}(P_2) = 2 * 3 > 5$$

so the $P_1 P_2$ is not a new class representative. Similarly, neither is $P_1 P_3$ or any multiples of P_2 or P_3. P_1^2 has a norm of 4 but $P_1^2 = \langle 2 \rangle$ which tells us the $[P_1]^2 \equiv 1$ and is not a new class. So there are no other ideals that represent new classes in the class group. Thus $h_K \leq 4$.

There is still the possibility that some of the prime ideals are actually principal ideals or that 2 or more of them are congruent to each other in the class group. If P were principal, $P = \langle a + b\sqrt{-14} \rangle$ for $a, b \in \mathbb{Z}$ which would imply that

$$\mathfrak{N}(P) = 2 = \mathfrak{N}(\langle a + b\sqrt{-14} \rangle) = N_{K/\mathbb{Q}}(a + b\sqrt{-14}) = a^2 + 14b^2 \implies a^2 = 2$$

which is clearly not possible since $a \in \mathbb{Z}$. Similarly, if P_2 or P_3 were principal then $3 = a^2$ which is not possible. So P_1, P_2, and P_3 are all non-principal ideals.

Now we need to ensure none of the prime ideals are congruent to each other in the class group. By way of contradiction say $P_1 \equiv P_2$. Then since $[P_1]^2 \equiv 1$, $[P_1]^{-1} \equiv [P_1]$ so

$$P_1 P_2 \equiv 1 \implies P_1 P_2 \equiv \langle a + b\sqrt{-14} \rangle \implies \mathfrak{N}(P_1)\mathfrak{N}(P_2) = 6 = a^2 + 14b^2 \implies a^2 = 6.$$

This clearly is a contradiction so P_1 is not congruent to P_2. By the same argument P_1 is not congruent to P_3. So this gives us at least three distinct classes: $[1], [P_1], [P_2]$.

We know $h_K \leq 4$. We also know $[P_1]^2 = 1$ which means $2 | h_K$ so $h_K = 2$ or 4. If $P_2 \equiv P_3$ then there would only be three unique classes which we know isn't true. So P_2 and P_3 must be distinct class representatives. Thus $Cl_K = \{[1], [P_1], [P_2], [P_3]\}$ and $h_K = 4$. So

$$Cl_K \cong \mathbb{Z}_4 \text{ or } \mathbb{Z}_2 \times \mathbb{Z}_2.$$

To determine which group the class group is isomorphic to we just need to know the order of $[P_2]$. If it's 2,

$$\mathfrak{N}(P_2^2) = a^2 + 14b^2 \implies a = \pm 3 \implies P_2^2 = \langle \pm 3 \rangle = \langle 3 \rangle = P_2 P_3.$$

But, $P_3 \neq P_2$ so this is a contradiction. Since the order of a class must divide the group order, the order of $[P_2]$ must be 4 which implies $Cl_K \cong \mathbb{Z}_4$ and $[P_2]$ is the generator of the class group.

5 Conclusion

This paper has established two foundational results in algebraic number theory. First, using the Minkowski embedding and the geometry of lattices, we proved that every ideal class in Cl_K contains a representative whose absolute norm is bounded by the Minkowski bound $M_K = \left(\frac{2}{\pi}\right)^s \sqrt{|d_K|}$. From this we concluded that h_K is finite. Second, we demonstrated the bound's computational utility by determining that $Cl_{\mathbb{Q}(\sqrt{-14})} \cong \mathbb{Z}_4$. The Minkowski bound makes the class group effectively computable. For imaginary quadratic fields $\mathbb{Q}(\sqrt{-d})$, the bound is small enough that class numbers can be determined by hand, as we demonstrated.

References

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